

Light Pollution Analytics and Celestial Visibility Planner

Alain Izabayo | Jeongho Lee | Jeffry Royce | Kristin Wiseman

Project Summary

The Light Pollution Analytics and Celestial Visibility Planner is a web application designed to help amateur astronomers, astrophotographers, and general stargazers make the most of their observation time by pairing environmental data with celestial tracking. The primary problem this application solves is the unpredictable nature of stargazing due to urban light pollution. Finding a "dark sky" location alone, however, doesn't address most users' structural knowledge gap regarding which specific stars, constellations, or deep-sky objects will actually be visible to the naked eye or through a lens from a given location or on a specific night.

As a solution, our application integrates two large-scale public datasets. The first dataset consists of geographic light pollution data, which maps limiting magnitudes to specific coordinates and ZIP codes. The second dataset is a comprehensive star catalog, containing tens of thousands of rows tracking celestial coordinates, brightness magnitudes, and object classifications. By joining these datasets, the application allows users to discover the best nearby observation points and instantly see an accurate, filtered list of visible celestial targets.

Creative Component

The technically challenging feature of this application is a Dynamic Visibility and Horizon-Line Calculator. Instead of simply displaying a static list of stars, the backend database will implement a filtering system that processes astronomical geometry based on user inputs. When a user queries a location, the application captures the exact latitude, longitude, and the current date and time. It then runs a multi-step relational query to mathematically calculate the altitude of celestial bodies relative to the user's horizon line.

Any star catalog object whose coordinates place it mathematically below the horizon line at that specific time is dynamically filtered out. Furthermore, the backend will cross-reference the user's local light pollution index against the celestial object's

apparent magnitude. If an object's brightness score is too faint to pierce the local light pollution threshold, it is automatically flagged as invisible. This ensures that the user is presented with a dynamically computed, mathematically rigorous checklist of what is overhead and visible in real time.

Beyond individual stars, the application also evaluates the visibility of entire constellations from a given location — a feature not found in existing stargazing or light pollution tools. When a user queries an area, the app determines how many of a constellation's member stars clear the local light pollution threshold and displays both the percentage of visible stars and a plain-language flag, such as "70% (Partially Visible)" or "90% (Fully Visible)." This prevents users from traveling to a site expecting to see a recognizable star pattern, only to find that most of its defining stars are washed out by skyglow.

Usefulness

This platform delivers targeted utility to three distinct user personas within the amateur science community:

- **Casual Stargazers:** Users looking for short-term recreational outings can input a target region to instantly receive a clear, jargon-free checklist of what celestial objects are visible to the naked eye, preventing wasted travel to heavily polluted or poorly timed observation sites.
- **Astrophotographers:** For technical hobbyists, imaging deep-sky objects requires narrow windows of absolute darkness. Astrophotographers can use the app as a dashboard for tracking what they've photographed, finding windows of opportunity in their area, or even planning when and where to see rare celestial objects.
- **Local Observers:** While official public datasets capture macro-level urban skyglow, they fail to track micro-level changes on the ground associated with urban sprawl. Through a reporting utility, users can submit these micro-level updates such as flagging a newly installed neighborhood streetlamp that compromises a previously dark area. These community updates alert nearby observers, transforming the platform into an organic, user-verified map of local viewing conditions.

Existing platforms like [LightPollutionMap](#) or specialized forecasting apps like [Clear Outside](#) address parts of this problem space, but they operate in isolation.

LightPollutionMap provides excellent macro-level satellite overlays, but it functions

purely as a passive viewing map and lacks any astronomical catalog integration. Conversely, weather or astronomy planners display visibility charts but expect the user to already know the light pollution constraints of their specific coordinates.

Our application bridges this gap by directly intersecting these siloed domains into a single relational ecosystem. [Stellarium](#) comes closest to providing full integration, but is geared more for virtual skywatching through a computer screen. Furthermore, while existing maps rely strictly on historical satellite passes that overlook hyper-local, real-time changes on the ground, our platform introduces a crowdsourced reporting utility. This allows community members to flag localized light-sprawl events dynamically, allowing our app to grow more accurate over time.

Realness

Our application uses three real public datasets to determine which celestial objects are realistically visible from a user's location.

Light Pollution Data

We use observational data from Globe at Night, a citizen-science initiative that collects measurements of night sky brightness from participants around the world.

Source: <https://globeatnight.org/maps-data/>

Format: CSV

Size: 13402 rows

Degree: 17 columns

Each observation records environmental conditions at a specific place and time. Important fields include latitude, longitude, local data and time, limiting magnitude, SQM (Sky Quality Meter) readings, cloud cover, country, and observer comments

This dataset captures how dark the sky is at a given location, which directly affects what objects can be seen

Celestial Data

We use the HYG Database, a public catalog combining data from the Hipparcos, Yale, Bright Star, and Gliese catalogs

Source: <https://www.astronexus.com/projects/hyg-details>

Format: CSV

Size: 119629 rows

Degree: 37 columns

Each row represents an individual star and contains astronomical information such as its name, Hipparcos identifier, right ascension, declination, apparent magnitude, spectral type, constellation, distance, and luminosity

This dataset captures which stars exist in the sky and their observable properties.

Deep Sky Object Data

We also use an NGC/Messier catalog, which contains information about non-stellar celestial objects.

Source: <https://github.com/lgbouma/obs/blob/master/data/catalogs/ngc.csv>

Format: CSV

Size: 7767 rows

Degree: 6

Each row represents a deep-sky object such as a galaxy, open cluster, globular cluster, planetary nebula, or emission nebula. Important fields include NGC ID, Messier ID, object type, visual magnitude, right ascension, declination, and descriptive comments.

This dataset captures astronomical objects beyond individual stars, enabling the application to support more advanced stargazing and astrophotography use cases.

Connection Between the Datasets

The two datasets are connected through the user's observation request.

When a user enters a ZIP code or location, the application will

1. Convert the user's location into latitude and longitude.
2. Find nearby Globe at Night observations to estimate local sky darkness.
3. Use the observation's LimitingMag value to determine the faintest visible object under those conditions.
4. Determine which stars in the HYG database and which deep-sky objects in the NGC catalog are above the horizon using the user's latitude, longitude, date, and time together with each object's right ascension and declination.

5. Filter stars using (mag <= LimitingMag)

Only celestial objects that are both above the horizon and bright enough to overcome local light pollution will be presented to the user.

This approach allows the application to serve both casual stargazers seeking visible stars and experienced observers searching for deep-sky targets.

Suitability for Stage 2

These datasets also support the requirements for Stage 2 because they naturally translate into entities within the database design. The Globe at Night dataset can be modeled as a LightPollutionObservation entity, the HYG dataset as a Star entity, and the NGC catalog as a DeepSkyObject entity. Additional application entities such as User, ObservationList, ObservationTarget, and CommunityReport allow the database to contain more than five entities while supporting multiple relationship types required by the rubric. For example, a single user can create multiple observation lists and submit multiple community reports, representing one-to-many relationships. Observation lists and celestial objects form a many-to-many relationship, since each list can contain multiple celestial objects and each celestial object can appear in the lists of many different users through the ObservationTarget entity. Therefore, the selected datasets are not only real and sufficiently large, but also suitable for satisfying the conceptual and logical database design requirements of Stage 2.

Functionality

Functional List

- *Create*: Users create an account and can add celestial objects to a "Custom Observation List" (targets they want to find).
- *Read*: Users search by ZIP code to find the closest dark-sky area, or search for a celestial object to see whether it's visible tonight.
- *Update*: Users submit community updates on saved locations (e.g., "A new stadium was built here, light pollution is worse now").
- *Delete*: Users delete objects from their target checklist once they successfully spot them.

Community Notes: Users will be able to Create, Read, Update, and Delete notes about transient conditions for locations like “Lighting for the county fair for the next week significantly reduces visibility”.

Other functionality:

- Keyword search: the user will be able to input the name of a celestial object to check if it's visible to them or to find locations where it's visible.
- The site will ask the user to share their device's location (as an alternative to searching by zip code).

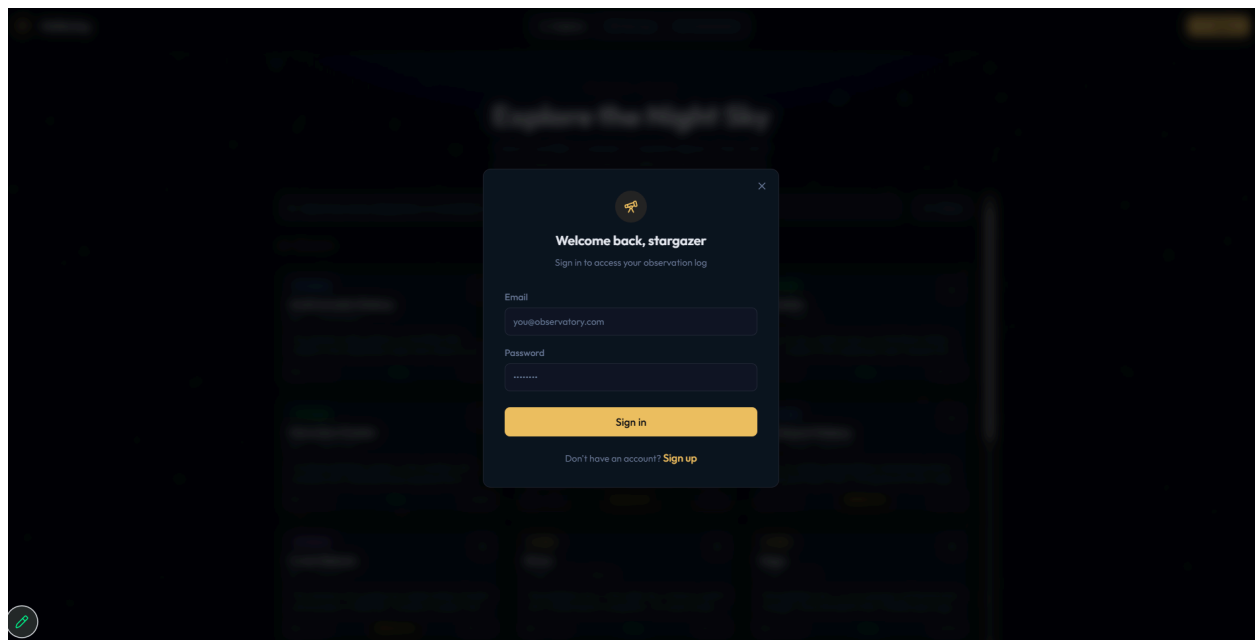
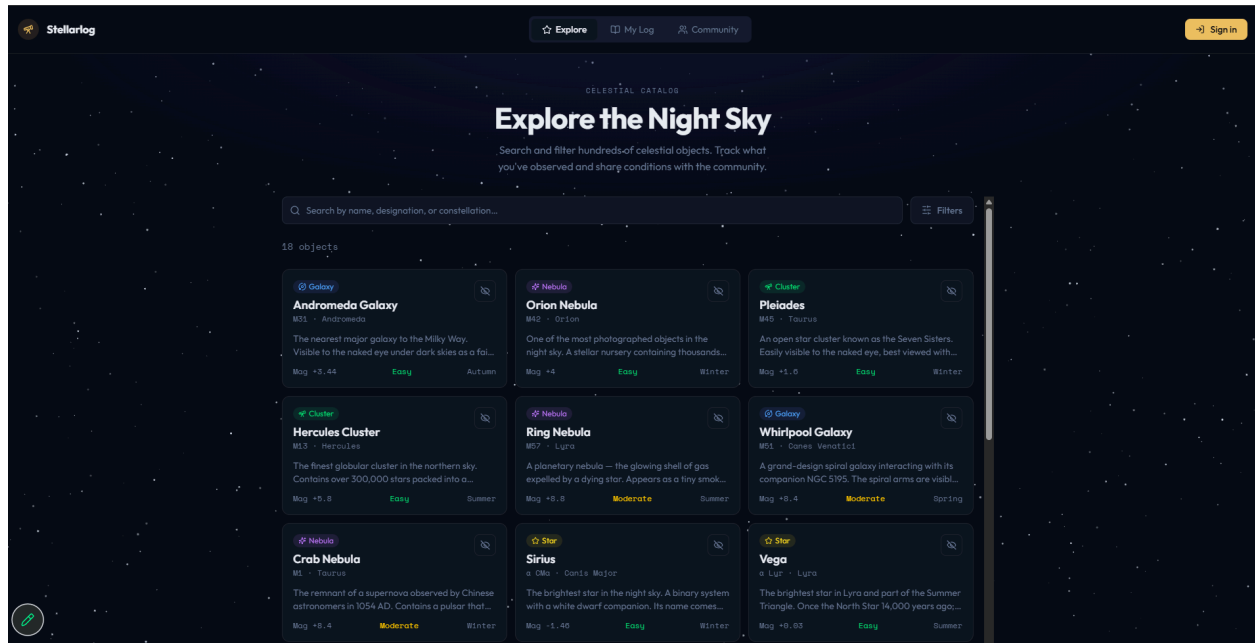
Work Distribution

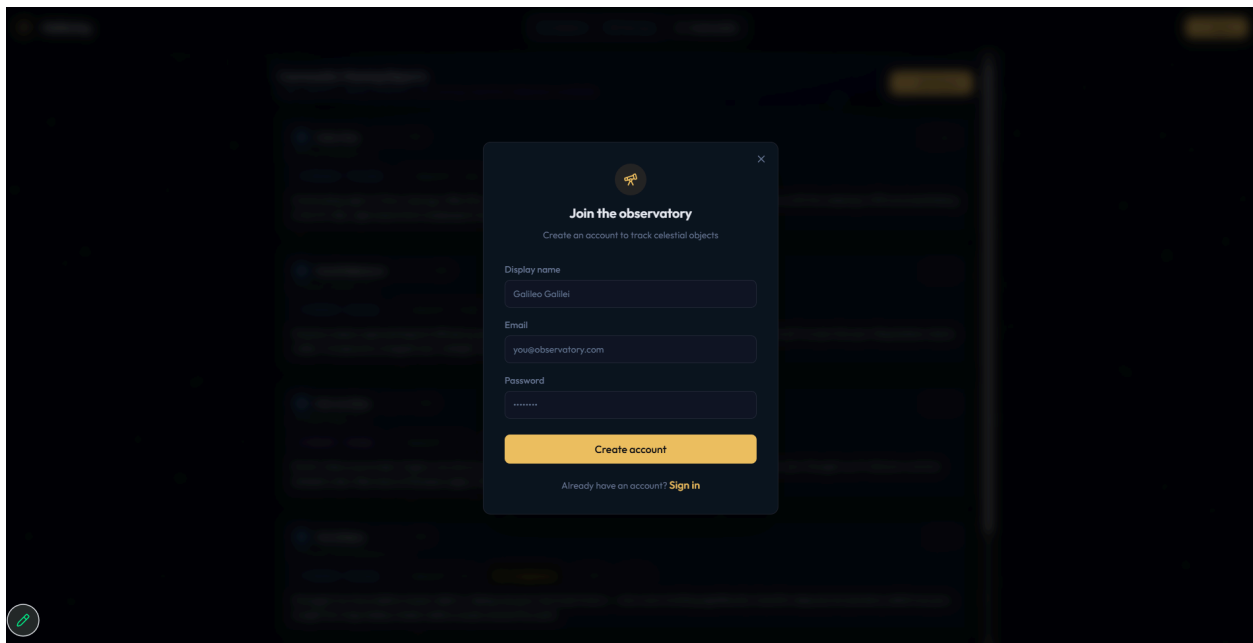
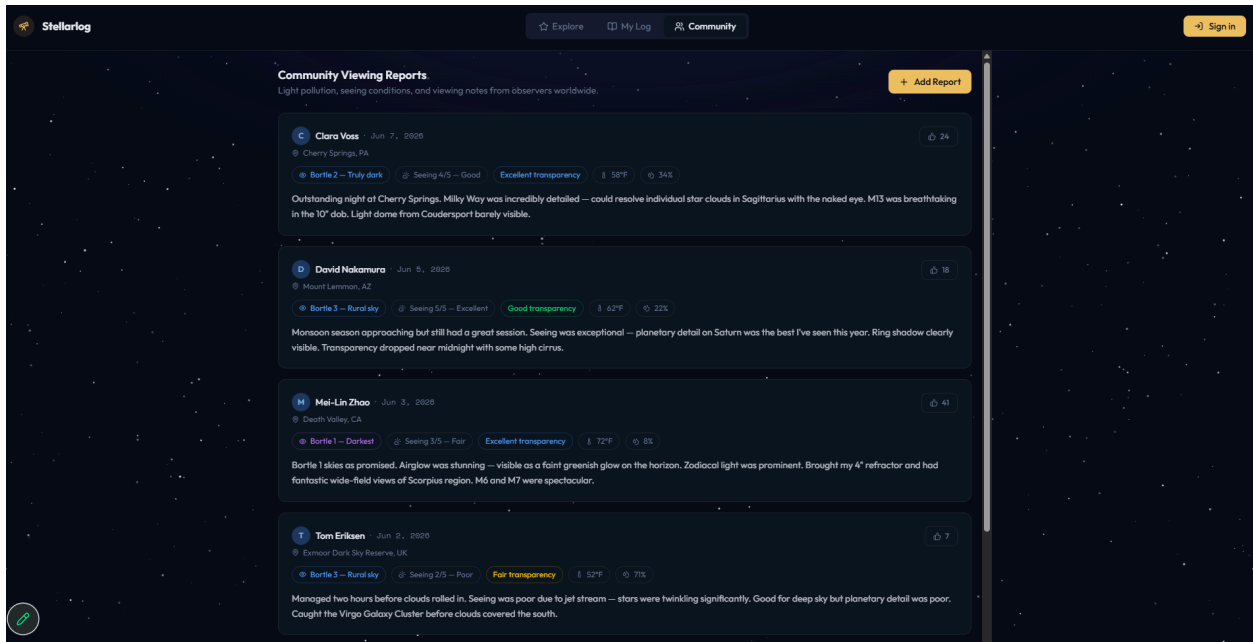
While the entire team shares responsibility for the overall development, integration, and submission of the platform, specific members will take the operational lead in the following core areas:

- Front End - Kristin
 - UI mockup design
 - Build layout templates and user-facing dashboards
 - Implementing visibility forms and navigation menus
- Back End - Jake
 - Add base data to database (e.g. the more “static” info about stars and other celestial objects)
 - Stored Procedure: Functional, with at least two advanced queries, cursors (optional), control structures, and application utility.
 - Trigger: Functioning triggers, involving an event, condition (IF statement), and action (Update, Insert, Delete), enhancing the application.
 - Conceptual/logical database design
 - Develop 3 advanced SQL queries
- Front/Back Communication - Jeffry
 - Manage data pipelines and connectivity between the frontend interface and backend systems
 - Develop routing to pass user coordinates and ZIP codes to the database
 - Implement the middleware logic for the dynamic horizon-line and constellation visibility formulas
- Logistics - Alain
 - Track key project deadlines and set reasonable internal deadlines
 - On standby to help any member who is struggling with meeting internal deadlines

- Coordinate team collaborative, internal communication, submission and TA-team relationships
- Assist with testing and QA across all components

UI Mockup





*UI Mockups generated with the help of [Figma Make](#) (AI based app prototyping tool).